

Machine Learning-Based Early Citation Prediction for AI and Climate Research

Project Proposal

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Project Idea

Artificial intelligence and climate-related research are increasing rapidly. However, it is difficult to identify at an early stage which research papers will become highly cited.

This project proposes a machine-learning system that will use early publication metadata from OpenAlex to predict whether an AI and climate-related paper will become a high-impact paper.

Problem Description

Citation impact usually becomes clear several years after publication. As a result, researchers and institutions cannot easily identify promising papers during their early publication period.

OpenAlex provides useful metadata such as publication year, authors, institutions, countries, publication venue, topic, open-access status, and citation count. These features may help predict the future citation impact of a paper.

Project Objective

The main objective is to classify research papers into two categories:

- High-impact paper: Top 10% of cited papers
- Not high-impact paper: Remaining papers

The project will also identify which metadata features have the greatest influence on citation impact.

Dataset

The dataset will be collected from the OpenAlex API.

- Research area: Artificial intelligence and climate or sustainability
- Publication years: 2020–2024
- Data type: Research-paper metadata
- Target label: High-impact or not high-impact

Proposed Methodology

The project will follow these main steps:

1. Collect research-paper metadata from OpenAlex.
2. Filter AI and climate-related papers.
3. Clean and process the data using PySpark.
4. Extract relevant features.
5. Create the high-impact target label.
6. Train and compare machine-learning models.
7. Evaluate the models and analyze feature importance.

The proposed models are Logistic Regression, Random Forest, and Gradient-Boosted Tree.

Evaluation

The models will be compared using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Expected Outcome

The expected outcome is a reproducible machine-learning pipeline that can predict potentially high-impact AI and climate research papers using early OpenAlex metadata.

The project may help researchers and institutions identify promising research earlier and understand the factors associated with higher citation impact.

Tools

Python, PySpark, OpenAlex API, and Jupyter Notebook or Google Colab will be used to complete the project.

References

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- [3] H. Zhu and L. Shuhuai, “Instant prediction of scientific paper cited potential based on semantic and metadata features,” *PLOS ONE*, 2024.